

Design and Implementation of a Pipelined Wallace Tree Multiplier using Verilog HDL

1. Cover Page

Project Title:

Design and Implementation of a Pipelined Wallace Tree Multiplier

Submitted by:

Name:

Register No:

Course:

Date:

2. Abstract

A multiplier is one of the most critical arithmetic units in digital systems such as processors, DSP processors, AI accelerators, and communication systems. The performance of a multiplier directly affects the overall system speed.

This project focuses on the design and implementation of a **pipelined Wallace Tree multiplier** using Verilog HDL. The Wallace Tree architecture reduces multiplication delay by using parallel partial product reduction through Full Adders and Half Adders.

Pipeline registers are introduced between different stages to improve the operating frequency and throughput of the multiplier.

The design is verified using a self-checking testbench with multiple random test cases.

3. Objectives

The objectives of this project are:

- Understand the Wallace Tree multiplier architecture.
 - Generate partial products using AND gates.
 - Implement partial product reduction using compressors.
 - Design pipeline stages for improved performance.
 - Develop Verilog RTL implementation.
 - Verify functionality using simulation.
 - Analyze latency and throughput.
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4. Design Specification

Parameter	Specification
Input Width	8-bit
Inputs	A[7:0], B[7:0]
Output	16-bit Product
Architecture	Wallace Tree
Pipeline	3/4 Stage Pipeline
Language	Verilog HDL
Reset	Active Low
Simulation Tool	VCS / ModelSim
Target Frequency	50 MHz

5. Wallace Tree Multiplier Theory

5.1 Basic Multiplication

5.2 Wallace Tree Reduction

6. Proposed Architecture

Block Diagram

7. Pipeline Architecture

Stage 1

Stage 2

Stage 3

8. RTL Design

Project directory:

Wallace_Multiplier/

```
|  
|-- rtl/  
| |  
| |-- wallace_top.v  
| |-- partial_product.v  
| |-- full_adder.v  
| |-- half_adder.v  
|  
|-- tb/  
| |  
| |-- wallace_tb.v  
|  
|-- sim/  
|  
|-- reports/
```

9. RTL Module Description

Full Adder Module

Function:

Adds three inputs.

Equation:

$$\text{Sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Carry} = AB + BC + AC$$

Partial Product Generator

Function:

Creates:

$A[i] \text{ AND } B[j]$

Example:

```
assign pp[i][j]=A[i]&B[j];
```

Wallace Reduction Module

Function:

Reduces:

64 bits

into:

2 rows

using:

- HA
 - FA
-

10. Testbench Description

The testbench performs:

- Clock generation
 - Reset sequence
 - Random input generation
 - Output checking
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Test Cases

Example:

Input:

A = 10

B = 20

Expected:

Product = 200

Random testing:

repeat(1000)

Generate random A,B

Compare:

RTL output

vs

A*B reference

11. Simulation Results

Include:

Waveform Screenshot

Show:

- clk
- reset
- input A
- input B
- output Product
- valid signal

Example:

Cycle:

- 1 Input applied
 - 2 Pipeline stage 1
 - 3 Pipeline stage 2
 - 4 Output valid
-

12. Verification Result

Example:

Test	Result
Basic multiplication	PASS
Random test	PASS
Maximum input	PASS
Zero input	PASS

13. Performance Analysis

Throughput

14. Comparison

15. Conclusion

16. Future Enhancements